THE CHARACTERISTIC POLYNOMIAL OF AN ELEMENT OF JR OVER A FINITE GROUP ACTION HAS COEFFICIENTS IN THE IDEAL J

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'A computation we omit shows that a_i in R^G'; that sentence is kept verbatim and is the one step the author leaves to the reader.

Allowed standard prerequisites (editorial).

- Invariant subring R^G of a ring with a finite group action. fixed by the statement at source lines 32570-32571 and used at proof line 32577. Fixes the target of the conclusion: the coefficients a_i must lie in R^G, not merely in R.
- The characteristic polynomial of an element and its expression as a product over the group. the explicit product prod_{sigma in G} (T - sigma(x)) in the statement at lines 32572-32573. Supplies the object whose coefficients a_i are being shown to lie in J.

Lemma 0.2. *Let G be a finite group of order n acting on a ring R. Let $J \subset R^G$ be an ideal. For $x \in JR$ we have $\prod_{\sigma \in G}(T - \sigma(x)) = T^n + a_1 T^{n-1} + \ldots + a_n$ with $a_i \in J$.*

Proof. Observe that the polynomial is indeed monic and has coefficients in R^G. We can write $x = f_1 b_1 + \ldots + f_m b_m$ with $f_j \in J$ and $b_j \in R$. Thus, arguing by induction on m, we may assume that $x = y - fb$ with $f \in J$, $b \in R$, and $y \in JR$ such that the result holds for y. Then we see that

$$\prod\nolimits_{\sigma \in G}(T - \sigma(x)) = \prod\nolimits_{\sigma \in G}(T - \sigma(y) + f\sigma(b)) = \prod\nolimits_{\sigma \in G}(T - \sigma(y)) + \sum_{i=1,\ldots,n} f^i a_i$$

where we have

$$a_i = \sum\nolimits_{S \subset G,\ |S|=i} \prod\nolimits_{\sigma \in S} \sigma(b) \prod\nolimits_{\sigma \notin S}(T - \sigma(y))$$

A computation we omit shows that $a_i \in R^G$ (hint: the given expression is symmetric). Thus the polynomial of the statement of the lemma for x is congruent modulo J to the polynomial for y and this proves the induction step. □

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